



RS003 Week 14 — English Worksheet

Topic: A HUD and Many Targets · **Course:** Pygame Turret · **Time:** about 45 minutes

This week your game shows a **score and timer** on screen (a HUD) and has **several targets** at once. Each target is a dictionary with its own position, size, and colour, and smaller targets are worth more points.

Words to know: **HUD**, **score**, **timer**, **list of dictionaries**, **render text**

1 · Predict 🧠

Read each piece of code. Before you run it, write what you think will happen.

```
font = pygame.font.SysFont(None, 32)
score_text = font.render(f"Score: {score}", True, (255, 255, 255))
screen.blit(score_text, (10, 10))
```

These three lines put text on screen. What does the player see in the top-left corner?


```
targets = []
for i in range(3):
    targets.append({"x": random.randint(150, WIDTH - 150), "y": HEIGHT - 200,
                    "size": random.choice([15, 25, 35])})
```

How many targets are made? What is different about each one?


```
score = score + (50 // hit["size"])
```

Score goes up by 50 // size . A small target has size 15, a big one has size 35. Which gives more points?

2 · Run It

 **Type the example, run it, and watch the score**

```

import pygame
import random
pygame.init()

WIDTH, HEIGHT = 800, 600
CENTER_X = WIDTH // 2
screen = pygame.display.set_mode((WIDTH, HEIGHT))
clock = pygame.time.Clock()
font = pygame.font.SysFont(None, 32)

targets = []
for i in range(3):
    targets.append({"x": random.randint(150, WIDTH - 150), "y": HEIGHT - 200,
"size": random.choice([15, 25, 35]), "color": random.choice([(255, 50, 50), (50,
255, 50), (255, 200, 50)])})

def draw_target(t):
    pygame.draw.circle(screen, (0, 0, 0), (t["x"], t["y"]), t["size"] + 3)
    pygame.draw.circle(screen, t["color"], (t["x"], t["y"]), t["size"])

def draw_hud(score, time_left):
    screen.blit(font.render(f"Score: {score}", True, (255, 255, 255)), (10, 10))
    screen.blit(font.render(f"Time: {time_left}", True, (255, 255, 255)), (WIDTH -
150, 10))

def get_aimed_target(gun_x):
    for t in targets:
        if abs(gun_x - t["x"]) < t["size"]:
            return t
    return None

gun_x = CENTER_X
score = 0
start_time = pygame.time.get_ticks()
GAME_DURATION = 30000

running = True
while running:
    elapsed = pygame.time.get_ticks() - start_time
    time_left = max(0, (GAME_DURATION - elapsed) // 1000)
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            running = False
        if event.type == pygame.KEYDOWN and event.key == pygame.K_SPACE:
            hit = get_aimed_target(gun_x)

```

```

        if hit:
            score = score + (50 // hit["size"])
            hit["x"] = random.randint(150, WIDTH - 150)
            hit["size"] = random.choice([15, 25, 35])

    keys = pygame.key.get_pressed()
    if keys[pygame.K_LEFT] and gun_x > 100:
        gun_x -= 6
    if keys[pygame.K_RIGHT] and gun_x < WIDTH - 100:
        gun_x += 6

    screen.fill((30, 30, 60))
    pygame.draw.rect(screen, (50, 80, 50), (0, HEIGHT - 50, WIDTH, 50))
    for t in targets:
        draw_target(t)
    pygame.draw.rect(screen, (0, 100, 0), (CENTER_X - 30, HEIGHT - 90, 60, 40))
    aimed = get_aimed_target(gun_x)
    marker_color = (255, 255, 0) if aimed else (200, 0, 0)
    pygame.draw.line(screen, (80, 80, 80), (CENTER_X, HEIGHT - 90), (gun_x, HEIGHT
- 200), 10)
    pygame.draw.circle(screen, marker_color, (gun_x, HEIGHT - 200), 8)
    draw_hud(score, time_left)
    pygame.display.flip()
    clock.tick(60)

pygame.quit()

```

Aim at the smallest target and press SPACE, then a big one. Which gave you more points? What does the timer do?

3 · Spot the Bug 🐛

Each block below was meant to do something but is broken. Fix it, then explain why the original was wrong.

Bug A — The score is supposed to appear on screen, but the number never shows up.

```
score_text = font.render(f"Score: {score}", True, (255, 255, 255))
```

Hint: rendering the text is not enough — it must be put on the screen.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — Hitting a small target should give more points than a big one, but every target gives the same score.

```
score = score + 5
```

Hint: the score should depend on the target's size.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — Pressing SPACE on empty space should do nothing, but the program crashes when you miss.

```
hit = get_aimed_target(gun_x)
score = score + (50 // hit["size"])
```

Hint: when nothing is aimed at, `hit` is `None`, and `None["size"]` is an error.

Write the fixed code:

Why was it wrong? Why does your fix work?

4 · Modify It

Add more targets

Change `range(3)` so more targets appear at once.

Write the new number you chose and how the game feels with more targets:

Change the game length

Change `GAME_DURATION` to make the round longer or shorter (it is in milliseconds, so 30000 is 30 seconds).

Write the new duration you chose and what 1000 milliseconds equals in seconds:

Hint: keep the targets inside `150` to `WIDTH - 150` so they never spawn off screen or under the HUD.

5 · Make It 📷

Build a scored shooting game

Build a program where:

1. several targets appear at once,
2. a HUD shows score and time,
3. smaller targets are worth more,
4. missing does not crash the game.

Send a **video** of 30 seconds of play, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I stored my targets as a list of ...

My HUD shows ... in the corner.

Small targets give more points because ...

I stopped misses from crashing by ...

One tricky moment was when ...

If I had more time, I would ...

6 · Record Your Walkthrough 🎥

Take a video of a full 30-second round. Talk through it like you are teaching someone. Try to use these words: **HUD**, **score**, **timer**, **dictionary**, **render**.

1. Show the HUD with score and time changing.
2. Hit a small target and a big one, and say the point difference.
3. Read the line that renders the score out loud.
4. Show what happens when you miss.

Write what you will say in your video. Plan it here before you record.

Submit 

Send this worksheet + your 30-second play video to teacher on KakaoTalk.